

Abstract:

Power electronic circuits are widely used in various applications such as power conversion, motor drives, renewable energy systems, and power supplies. Numerical simulations play a crucial role in the design, analysis, and optimization of these circuits. This report presents an overview of the key aspects involved in the numerical simulation of power electronic circuits. The fundamental principles of power electronics, including switching devices, converter topologies, and control strategies, are discussed. The importance of accurate modeling and simulation techniques for predicting the behavior of power electronic circuits is emphasized. The report also highlights the challenges and limitations associated with numerical simulations and provides insights into the future trends and research directions in this field. The presented information can serve as a valuable resource for researchers, engineers, and students working in the area of power electronics and its applications.